MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPES 2J49 AND 2J50

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed; fixed frequency (2J49) 9,080 ± 80 MHz, (2J50) 8,825 ± 75 MHz; magnet separate; 50-kw rated peak power output.

ABSOLUTE RATINGS:

Parameter:	Ef	tk	tpc	Du	epy	Pi	pi	ib	T(anode)	Alt
Unit:	V	sec	µs	---	kv	W	kw	a	°C	ft
Maximum:	7.0	---	2.5	0.0012	16	180	260	16	100	10,000
Minimum:	--- (Note 2)	120	---	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Marking: See note 3
(D) Weight: 23 ounces (652.04 grams) approximately
Cathode: Unipotential

TEST CONDITIONS:

Parameter:	Ef	tk	tpc	Du	Ib	VSWR	H
Unit:	V	sec	µs	---	mAdc	Ratio	Gauss
Tolerance:	---	---	±10%	---	---	---	---
Test 1:	6.3	120(max)	---	---	---	---	---
Test 2:	0	---	1.0	0.001	12	1.05:1	5,400
Test 3:	0	---	1.0	0.001	11	1.5:1	5,400
Test 4:	0 (Note 2)	---	1.0	0.001	13	1.5:1	5,400 (Note 4)

GENERAL:

Qualification - Required

(D) denotes changes

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	METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
							MINIMUM	MAXIMUM	
		Qualification inspection							
(D)	1002	Barometric pressure, reduced	-	2	Pressure = 380 mmHg (absolute)	---	---	---	---
(D)	4027	Temperature coefficient	-	2		ΔF/°C	---	0.25	MHz
(D)	1047	Low-temperature operation	-	2	tk = 120 sec (max)	---	---	---	---
(D)	1031	High-frequency vibration	-	-	No voltages applied	---	---	---	---
		Quality conformance inspection, part 1	1	-					
	4003	Pressurizing	-	-	40 to 45 psia	---	---	---	---
(D)	4289	Heater current	-	1		If	0.90	1.10	A
	4223	Fixed-tuned frequency	-	2	T(anode) = 80° +10°C Type 2J49 Type 2J50	 F F	 9,000 8,750	 9,160 8,900	 MHz MHz
	4306	Pulse voltage	-	2		epy	11	13	kv
	4250	Power output	-	2	t = 300 (max)	Po	40	---	W
	4308	RF bandwidth	5	3,4		BW	---	2.5	MHz
	4310	Pulling factor	-	2		ΔF	---	15	MHz
(D)	4315	Stability	6	2		MP	---	0.5	%
		Quality conformance inspection, part 2							
	1031	Low-frequency vibration	-	-	No voltages applied	---	---	---	---
		Quality conformance inspection, part 3							
	---	Life test	7	-	Group D	Life	600	---	Cycles
	---	Life-test end points:							
	4250	Power output	-	2		Po	30	---	W
	4308	RF bandwidth	-	2		BW	---	4	MHz

NOTES:

1. Unless otherwise specified the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

2. During high voltage pulsed operation, the heater should be varied with input power in accordance with the following formula: $Ef = 6.3\sqrt{1 - \frac{Pi}{150}}$ with Ff = 0 for all values of Pi greater than 150.

3. In addition to the regular markings, the frequency of each 2J50 tube shall be indicated by three colored dots in accordance with the following color code:

   Orange, violet, green - - - - - - - - 8,750 to 8,800 MHz
   Orange, violet, blue - - - - - - - - 8,800 to 8,850 MHz
   Orange, violet, violet - - - - - - - - 8,850 to 8,900 MHz

   NOTE: The dots shall be 1/8 to 3/16 inch (3.18 to 4.76 mm) in diameter and placed in a row after the type marking.

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NOTES: -Continued

4. Field strength must be as specified with the magnet at operating temperature. Field strength to be measured in accordance with method 1367.

5. Bandwidth measurements shall be made at each anode current value into a 1.5 VSWR transmission line adjusted in phase to produce maximum spectrum degradation.

6. Stability shall be measured with the tube operating into a transmission line with a VSWR of 1.5:1 adjusted in phase to produce maximum instability. The missing pulses shall be counted during any 3-minute period of a test interval not to exceed 6 minutes.

(D) 7. VSWR shall be 1.5:1 minimum with phase varying through a minimum of 1/2 λ approximately every 15 minutes. Each cycle shall be conducted in accordance with the following schedule:

Test condition 1 - - - - - - - - -	2 minutes
Test condition 2 - - - - - - - - -	50 minutes
Off- - - - - - - - - - - - - - - -	8 minutes

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2948)

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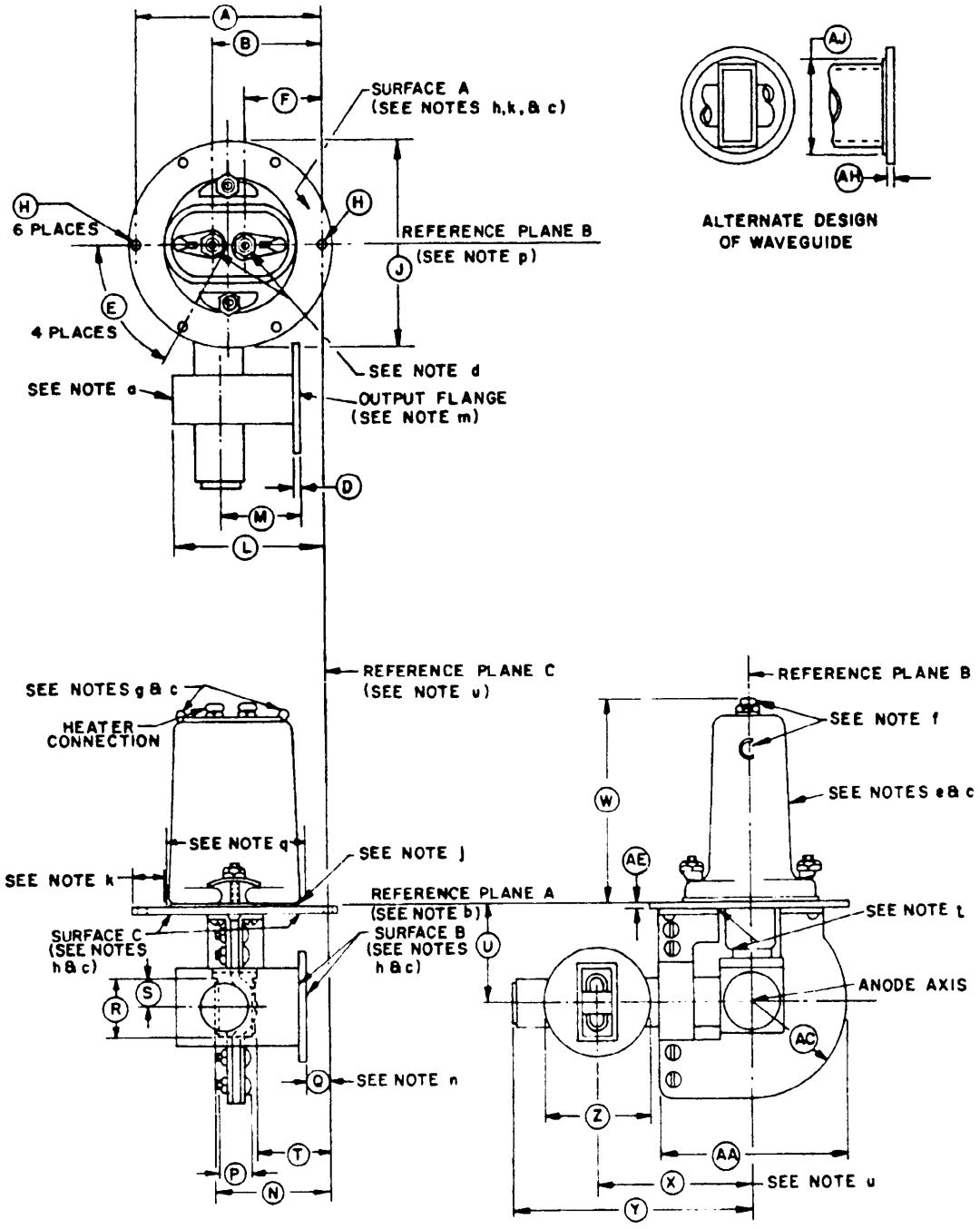


FIGURE 1. Outline drawings of electron tube types 2J49 and 2J50.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Qualification inspection			
E	59°48'	60°12'	
Quality conformance inspection, part 1			1
B	1.664 (42.27)	1.710 (43.43)	
D	.080 (2.03)	.090 (2.29)	
F	1.164 (29.57)	1.210 (30.73)	
H	.190 (4.83)	.196 (4.98)	
N		1.750 (44.45)	
P		.500 (12.70)	
Q	.417 (10.59)	.457 (11.61)	n
T	1.125 (28.58)		
U	1.542 (39.17)	1.582 (40.18)	
X	2.417 (41.39)	2.457 (42.41)	
Z	1.743 (44.27)	1.757 (44.63)	
AH	.080 (2.03)	.090 (2.29)	
AJ		1.446 (36.73)	
Quality conformance inspection, part 2			
A	2.869 (52.87)	2.881 (53.18)	
J	3.219 (81.76)	3.281 (83.34)	
L		2.625 (46.68)	
M		1.187 (30.15)	
R	.969 (24.61)	1.031 (26.19)	
S	.469 (11.91)	.531 (13.49)	
W	2.922 (54.22)	3.047 (77.39)	
Y		3.906 (99.21)	
AA		3.125 (79.38)	
AC		1.500 (38.10) R	
AF	.120 (3.05)	.130 (3.30)	

FIGURE 1. Outline drawing of electron tube types 2J49 and 2J50 - Continued.

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NOTES:
a. Section of waveguide RG-51/U.
b. Reference plane A is defined as that plane which coincides with the face of the mounting plate.
c. Qualification inspection applies.
d. Jacks shall be locking type as illustrated in MIL-E-1. The centerline of the jack holes shall be within .023 (.58 mm) inch of reference plane B, and shall be perpendicular, within 3° of reference plane A.
e. Insulator shall be pyrex glass, porcelain, or approved equivalent.
f. Common heater-cathode connection shall be identified by the letter C.
g. Filament leads shall be flexible and slack, and shall be soldered to terminal clips.
h. All metal surfaces shall be painted with heat resistant, noncorrosive paint except surfaces A, B and C which shall be free from paint.
j. Asbestos gasket is optional.
k. With surface A resting on a plane surface, flatness of mounting plate .500 (12.70 mm) inch from outer edge shall be such that a gage .010 (.25 mm) inch thick and .125 (3.18 mm) inch wide will not enter more than .250 (6.35 mm) inch. Quality conformance inspection, part 2 applies.
m. Protective cover shall be provided for output flange.
n. Tolerances include angular and lateral deviations.
p. Reference plane B is defined as that plane which passes through the centers of holes H and H_1 and is perpendicular to reference plane A.
q. Any portion of the assembly within this area shall lie within a 1.109 (28.17 mm) inch radius of the true center of the mounting plate.
r. Output flange and open end of waveguide shall be concentric within .010 (.25 mm) inch.
s. Waveguide assembly shall provide hermetic seal with output flange.
t. Coaxial tube shall provide hermetic seal with surface A.
u. Reference plane C is defined as that plane which passes through the center of hole H_1 and is mutually perpendicular to reference planes A and B.

FIGURE 1. Outline drawing of electron tube types 2J49 and 2J50 - Continued.

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